ONOTE: Benchmarking Omnimodal Notation Processing for Expert-level Music Intelligence

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Appendix

A ONOTE Implementation Details

The ONOTE benchmark is constructed over three music notation systems: standard staff notation, Jianpu, and guitar tablature, to support unified evaluation for omnimodal music score understanding. The dataset is built through a rigorous and controllable three-stage pipeline. After initial data collection, strict cross-modal alignment and format conversion are performed to ensure notation diversity and test validity. Based on the aligned omnimodal and cross-format data, task-specific test samples are constructed for the four core tasks of ONOTE. All samples are manually verified to guarantee musical correctness and structural validity. In total, the benchmark consists of 1,120 high-quality test samples.

A.1 Visual Score Understanding (VSU) Data

For the Visual Score Understanding task, we design one specialized visual question-answering (VQA) question for each music score image. All VQA questions are rigorously examined, revised, and validated by professional music annotators to ensure clarity, accuracy, and alignment with music-theoretic foundations. Each test sample in this task is composed of a single music score image in one of the three formats—standard staff notation, Jianpu, or guitar tablature—paired with its corresponding professionally designed VQA question. During evaluation, the model's predicted answer is automatically compared with the authoritative ground-truth answer, and classification accuracy is computed to quantify performance. Through this process, we objectively assess the model's ability of visual perception, symbol recognition, and high-level music-theoretic understanding directly from music score images without additional textual guidance.

A.2 Cross-Format Notation Conversion (CNC) Data

The Cross-Format Notation Conversion task focuses on assessing the model's ability to translate music content across different representation systems while preserving musical semantics. For every input music score image in one notation format, we prepare a complete, semantically equivalent symbolic text in another target notation as the standardized ground truth. In particular, we provide standard staff notation texts as references for both guitar tablature and Jianpu inputs, and supply standard Jianpu texts for staff notation inputs to form comprehensive cross-format conversion pairs. When the model receives a source score image as input, it generates a symbolic sequence in the target notation, which is then automatically compared with the pre-prepared ground-truth text at the level of music logic, pitch consistency, rhythm accuracy, and structural completeness. This comparison allows us to quantitatively evaluate the model's genuine music-theoretic reasoning and cross-format mapping capabilities.

A.3 Audio-to-Symbolic Transcription (AST) Data

In the Audio-to-Symbolic Transcription task, each music score image in the dataset is strictly aligned and synchronized with a corresponding high-quality music audio clip to ensure consistent musical content across modalities. In addition to aligned audio data, we also prepare complete, standardized symbolic ground-truth texts in the target notation format for each audio clip. During evaluation, the model takes a segmented music audio as input and predicts the corresponding symbolic music notation. The model-generated transcription is then quantitatively compared with the reference text from multiple dimensions, including pitch sequence, temporal position, note duration, and structural organization. By measuring the consistency between the predicted sequence and the ground truth, we reliably evaluate the model's audio perception ability, temporal alignment precision, and the accuracy of end-to-end audio-to-symbolic transcription.

A.4 Symbolic Music Generation & Aesthetics (SMG) Data

Different from the previous three discriminative and transductive tasks that rely on pre-collected score images or audio, the Symbolic Music Generation & Aesthetics task is a generative task that evaluates the model's capacity for creative music expression. This task does not use existing score images as input; instead, it examines whether the model can independently produce coherent, standardized, and aesthetically reasonable symbolic music under given constraints. During evaluation, we focus on three core aspects: the overall quality of generated music, the rationality of music structure, and the continuity of musical aesthetics. By assessing formal correctness, structural logic, and artistic expression of the generated notation, we comprehensively evaluate the model's higher-level music creation ability beyond understanding and transcription.

B Prompt used in ONOTE

B.1 Task Execution Prompts

In this section, we detail the exact execution prompts employed to evaluate the capabilities of Omni-models across the four primary tasks of the ONOTE benchmark.

To ensure a fair evaluation, we adopted a standardized prompting paradigm. Each prompt is structured with a specific system **Role** (to activate the relevant domain expertise of the LLM) and a precise **Task** instruction.

Furthermore, to systematically prevent model hallucination and ensure standardized output parsing during the automated evaluation, we applied a strict set of global formatting constraints (e.g., specific JSON schemas, sequence length limits, and anti-loop rules). To focus on the core semantic requirements of each task, these global formatting constraints are omitted from the individual task descriptions below.

The core semantic prompts for the four primary evaluation tracks are illustrated in Figure 6.

In evaluating the capabilities of Large Language Models (LLMs) across diverse music notation systems, unstandardized outputs often lead to the failure of automated parsing scripts. To ensure the fairness of the evaluation and the computability of the results, ONOTE introduces a strict set of **Global Formatting Constraints**.

Independent of the task-specific prompts, these global constraints explicitly define the standard output specifications for three core music notation representations: ABC Notation, Numbered Musical

ONOTE Task Execution Prompts

Task 1: Symbolic Music Generation

Role: You are a top-tier composer.
Task: Please compose an original classical-style melody for me.
Key Signature: C Major. Time Signature: 4/4. Length: Strictly 8 measures.

Task 2: Cross-Format Notation Conversion

Role: You are taking a <notation>exam.
Task: Please answer by outputting ONLY the single letter of the correct option (A, B, C, or D).

Task 3: Visual Score Understanding

Role: You are a highly precise Music OCR and Transcription AI.
Task: Convert the provided sheet music image into Numbered Musical Notation (Jianpu).

Task 4: Audio-to-Symbolic Transcription

Role: You are an extremely precise AI music transcription expert.
Task: Please transcribe the first 10 seconds of this audio.
Transcribe the sequence exactly as you hear it.
Do not repeat the same note indefinitely. If you are unsure, stop transcribing.
Transcribe no more than 60 notes for these 10 seconds.

Figure 6: Core semantic execution prompts for the four primary tasks in the ONOTE benchmark

Notation (Jianpu), and Structured Pitch-Duration Sequences. By enforcing these rules in the system instructions, we effectively mitigate formatting hallucinations and ensure that the models focus purely on the semantic music tasks.

The detailed definitions of these formatting standards are presented in Figure 7.

B.2 Scoring Assessment Prompts

To enable a scalable, multi-dimensional, and nuanced evaluation of the generated music notations, ONOTE employs an *LLM-as-a-Judge* paradigm. While traditional string-matching metrics fall short in assessing the musicality and structural validity of complex scores, state-of-the-art LLMs can effectively serve as automated expert evaluators when guided by rigorous rubrics.

In this subsection, we detail the **Scoring Assessment Prompts** designed for our evaluation pipeline. To ensure objectivity and consistency, each critic prompt is meticulously engineered with three core components: 1) **Expert Persona:** Assigning a strict and domain-specific role (e.g., a rigorous music theory professor or a fingerstyle guitar master) to calibrate the model's evaluation standards. 2) **Objective Verification Constraints:** Mandating the mathematical verification of rhythmic values (e.g., exactly 4.0 beats per measure) and structural layouts before assessing subjective aesthetics. 3) **Standardized Output Format:** Forcing the model to output a parsable, continuous score (ranging from 1 to 5) without generating redundant conversational text.

The detailed critic prompts corresponding to the three primary notation formats (ABC Notation, Jianpu, and ASCII Guitar Tablature) are presented in Figure 8.

C ONOTE Evaluation Metrics Implementation Details

C.1 Overall Evaluation Framework

The ONOTE benchmark designs independent automated evaluation logics for four distinct tasks, all implemented through deterministic rules and sequence matching algorithms without subjective scoring or LLM-as-a-judge, ensuring reproducibility and impartiality. The evaluation process first performs structural parsing of model outputs, then executes task-specific verification logic, and finally provides quantitative results from dimensions including pitch accuracy, temporal consistency, music-theoretic compliance, and structural integrity.

C.2 Canonical Pitch Space Projection

Prior to task-specific evaluation, to achieve fair comparison across notation systems, all model-generated notation outputs are uniformly mapped to a one-dimensional temporally ordered canonical pitch sequence. The specific processing logic is as follows: for guitar tablature, the symbolic string-fret correspondences are accurately converted to standardized MIDI pitches by combining string position, fret information, and standard tuning rules; for Jianpu (numbered musical notation), the corresponding absolute scientific

Global Constraints: Music Notation Formatting Standards

System Instruction: When generating or transcribing music, you must strictly adhere to one of the following three designated notation formats as requested in your specific task. Do not mix formats.

Format A: ABC Notation

1. **Metadata:** Must include standard headers (e.g., `X:1`, `T:Title`, `M:4/4`, `K:C`).
2. **Syntax:** Use standard ASCII characters for pitches, accidentals, and rhythms.
3. **Structure:** Group notes strictly according to the time signature. Use `|` to separate measures.

Format B: Numbered Musical Notation (Jianpu)

1. **Pitch:** Use numbers 1-7 for scale degrees, and 0 for rests.
2. **Octave Modifiers:** Prefix with _ for one octave lower (e.g., _5) and ˆ for one octave higher (e.g., ˆ1).
3. **Duration:** Append duration fractions in parentheses immediately after the pitch. Example: 3(1/4).
4. **Separation:** Use a single space to separate individual notes and `|` for bar lines.

Format C: Structured Pitch-Duration Sequence

1. **Syntax Form:** <Pitch><Octave>(<Duration>). Example: C4(1/4) or F#5(1/8).
2. **Separation:** Output must be a strictly space-separated sequence.
3. **Constraint:** Output ONLY the note sequence. Do not include measure lines, conversational text, or markdown wrappers.

Figure 7: The three standard music notation formatting constraints applied in ONOTE

pitches are calculated based on key signature, scale degree markings, and octave indicators; for chord notes within the same time frame, they are sorted uniformly from lowest to highest pitch to eliminate evaluation biases caused by differences in note ordering across notation systems, laying a unified pitch representation foundation for subsequent accurate task-specific evaluation.

C.3 Task-Specific Evaluation Implementation Details

Visual Score Understanding (VSU) Evaluation: As a classification task, VSU adopts a ground-truth answer matching mechanism. The specific process involves parsing the model-generated answer, conducting dual-level matching (string-level and semantic-level) with manually annotated ground-truth answers, determining correctness based on exact consistency, and calculating the overall classification accuracy. This evaluation focuses solely on the correctness of visual recognition and music-theoretic understanding, excluding assessments of sequence generation quality or format standardization.

Cross-Format Notation Conversion (CNC) Evaluation: As a sequence conversion task, CNC employs a combined evaluation approach of music-theoretic equivalence verification and sequence alignment. First, both the input notation and the model-generated output notation are projected to a unified canonical pitch sequence. Subsequently, the edit distance algorithm is used to calculate the matching accuracy of pitch sequences and rhythm sequences. Meanwhile, key signature, time signature, and overall structure are checked for consistency to ensure music-theoretic equivalence. Finally, the grammatical legality of the target format is verified, and outputs with format errors or unparseable content are directly rejected.

Audio-to-Symbolic Transcription (AST) Evaluation: As a temporal audio transcription task, AST adopts a dual-dimensional alignment evaluation logic (pitch and temporal alignment). First, the canonical pitch sequence, note onset times, and duration ground truths are extracted from the audio references. Then, the model-generated symbolic notation is converted into a sequence aligned with a uniform time axis, which is dynamically aligned with the ground-truth sequence. Pitch matching accuracy and temporal alignment accuracy are calculated separately. Outputs that are unparseable, contain illegal symbolic representations, or have severely mismatched lengths with the ground truth are directly deemed invalid.

Symbolic Music Generation & Aesthetics (SMG) Evaluation: As a generative task, SMG adopts a multi-level rule-based verification evaluation scheme. First, renderability verification is performed to check the grammatical completeness, symbolic legality, and format standardization of the generated notation, ensuring it can be normally parsed by mainstream music score engines. Second, music-theoretic and structural verification is conducted to validate the consistency of key signatures, the rationality of measure division, and the legality of rest and note representations. Finally, structural coherence evaluation is implemented to measure the completeness of musical phrases, the self-consistency of rhythmic logic, and the rationality of musical structure based on preset sequence rules.

LLM-as-a-Judge: Evaluation Prompts for Automated Scoring

Critic A: ABC Notation Evaluation

Role: As an extremely strict music theory professor, review and score AI-generated text-based ABC notation. Evaluate the model's proficiency by analyzing rhythmic values and musicality (Score 1-5).

1. Rhythmic Verification: A correct score contains exactly 4.0 beats per measure. The sum of rhythmic values between two `|` symbols must equal 4.0. (*Score 5: All correct; Score 3: Half correct; Score 1: All incorrect*).

2. Aesthetic Analysis: Motif development, melodic contour, and rhythmic groove should be rich and logical. (*Score 5: Complete motif and beautiful melody; Score 3: Discernible motif; Score 1: Lacks motif, unappealing*).

Output Constraints: Output ONLY the final scores without analysis. Format: `Technical Score: [ ]/5, Aesthetic Score: [ ]/5, Average Score: [ ]/5`.

Critic B: Numbered Musical Notation (Jianpu) Evaluation

Role: As an extremely strict music theory professor, review and score AI-generated Jianpu based on rhythmic values and musicality (Score 1-5).

Syntax Guide: `|` separates measures, spaces separate complete beats. A standalone number (e.g., 1, 0) or an extension dash (`-`) occupies 1 beat. Two numbers in parentheses (e.g., `(3 4)`) represent two eighth notes occupying 1 beat.

1. Rhythmic Verification: Exactly 4.0 beats per measure between `|` symbols. (*Score 5: All correct; Score 3: Half correct; Score 1: All incorrect*).

2. Aesthetic Analysis: Evaluate motif, melodic contour (rise/fall of notation numbers), and rhythmic groove. (*Score 5: Rich and logical; Score 1: Does not constitute actual music*).

Output Constraints: Output ONLY the scores without analysis. Format: `Technical Score: [ ]/5, Aesthetic Score: [ ]/5, Average Score: [ ]/5`.

Critic C: ASCII Guitar Tablature Evaluation

Role: As an extremely strict fingerstyle guitar master, review and score AI-generated ASCII guitar tabs based on layout, fingering, and musicality (Score 1-5).

1. Layout and Timing: The 6 strings must align perfectly vertically. Assuming 4/4 time, characters on each string between two bar lines (`|`) must be identical, exactly 16 characters per measure. (*Score 5: Layout correct, exactly 16 chars/measure; Score 3: Partial errors; Score 1: Mostly incorrect*).

2. Musicality Analysis: Voicing should have distinct layers (bass root, middle harmony, high melody). (*Score 5: Rich range/rhythm; Score 3: Monotonous; Score 1: Illogical*).

3. Fingering Feasibility: Analyze simultaneous notes in the same column. A reasonable stretch is a maximum span of 7 frets. (*Score 5: Logical, rarely ¿7 frets; Score 3: Frequent unreasonable stretches; Score 1: Physically impossible*).

Output Constraints: Output ONLY the scores without analysis. Format: `Technical Layout Score: [ ]/5, Fingering Score: [ ]/5, Musicality Score: [ ]/5, Average Score: [ ]/5`.

Figure 8: System prompts designed for the LLM-as-a-Judge automated evaluation, detailing strict scoring criteria for rhythmic verification, structural layout, and musical aesthetics across three notation formats